

Negative LDO (LT3093): Working Document

Digital Audio Player • Rev 10

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1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LT3093EDD#TRPBF (AL-PN #1-052), U203
Package	12-lead DFN, 3 × 3 mm, exposed pad (pin 13) = IN
Function	Ultralow-noise negative LDO, -3V9 → -3.3 V analog rail (-3V3)

2. Design Requirements

Requirement	Value
Input	-3V9 from the LT3462A inverter, -3.816 to -3.971 V
Output	-3.32 V
Load	I/V, summing, OPA1622 (V- side)
Output current	≈ 27 mA quiescent; ≈ 98 mA peak at 0.6 V RMS (200 mA rated)
Current limit	126.6 mA programmed (R_ILIM 15.4 kΩ)
UVLO	-3.5 V rising
Power good	-3.0 V (90.4 % of V_OUT), to GPIO11 (PG_-3V3)
VIOC	Unused, floated

Requirement	Value
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	–3V9, C_IN
3	EN/UV	R_EN2 from IN, R_EN1 to GND
4	PG	R_PU 100 kΩ to +3V3_DIG, GPIO11
5	PGFB	R_PG1 / R_PG2 midpoint
6	ILIM	R_ILIM to GND (Kelvin to pin 9)
7	VIOC	Not connected (float)
8	SET	R_SET C_SET to GND; C_SET cathode to SET (SET is at –3.32 V)
9	GND	GND
10	OUTS	OUT at the load
11, 12	OUT	–3V3, C_OUT, R_PG1
13	Exposed pad	IN (–3V9): own pour and via field, never GND

Source: [1], [2]

4. External Components

Ref	Value / Rating	MPN	AL-PN
R_SET (R232)	33.2 kΩ 0.1 % 0402	YAGEO RT0402BRD0733K2L	#1-085
C_SET (C222)	2.2 μF tantalum, 16 V, 0603, anode to GND	Vishay 298D225X0016M2T	#1-098
C_IN, C_OUT (C218, C216)	10 μF 10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R_EN1 (R226)	50 kΩ 0.1 % 0603	YAGEO RT0603BRD0750KL	#1-075
R_EN2 (R219)	88.7 kΩ 1 % 0402	YAGEO RC0402FR-0788K7L	#1-087
R_PG1 (R220)	135 kΩ 0.1 % 0603	YAGEO RT0603BRD07135KL	#1-086
R_PG2 (R224)	15 kΩ 0.1 % 0402	YAGEO RT0402BRD0715KL	#1-078
R_ILIM (R230)	15.4 kΩ 0.1 % 0402	YAGEO RT0402BRD0715K4L	#1-088
R_PU (R221)	100 kΩ 0.1 % 0402	YAGEO RT0402BRD07100KL	#1-131

Source: [2], [3]

5. Equations and Results

$$V_{OUT} = -I_{SET}R_{SET} = -100\mu A \times 33.2k\Omega = -3.32V (-3.254V \text{ to } -3.386V) \quad (1)$$

$$V_{IN(UVLO)} = -1.26(1 + 88.7/50) = -3.495V; \text{ falling: } -(1.26 - 0.215)(1 + 88.7/50) = -2.90V \quad (2)$$

$$V_{OUT(PG)} = -0.3(1 + 135/15) = -3.000V (90.4\%) \quad (3)$$

$$I_{LIM} = 1.95k\Omega \cdot A / 15.4k\Omega = 126.6mA \quad (4)$$

$$I_{PU} = 3.3V / 100k\Omega = 33\mu A (\leq 100\mu A \text{ PG spec}) \quad (5)$$

$$|V_{IN(MIN)}| - |V_{OUT(MIN)}| - V_{DO(MAX)} = 3.816 - 3.287 - 0.28 = 0.25V \quad (6)$$

Source: [1]

Test (simulated)	Result
Dropout (29 Ω / 16.5 Ω load)	203 mV at $\approx$ 115 mA; 214 mV at $\approx$ 201 mA
Noise vs C_SET (0.47 / 1 / 2.2 / 4.7 μ F)	1.44 / 0.75 / 0.44 / 0.34 μ V RMS
Output impedance	0.26 m Ω at 1 kHz; 5.2 m Ω at 20 kHz; 76 m Ω peak at 282 kHz
Start-up	Regulates at -3.62 V input; PG at -3.000 V after 3.9 ms; settled 166 ms after PG
Brownout	Loses regulation at -3.490 V; PG low at -3.129 V; off at -2.957 V
Load step 27 $\leftrightarrow$ 205 mA	-7.6 / +3.6 mV, settles in $\approx$ 20 μ s
Input ripple 7.7 mV pp at 1.2 MHz	1.67 μ V pp at OUT

6. Important Values

Item	Value
EN/UV threshold	-1.26 V (-1.20 to -1.33 V), 215 mV hysteresis
PGFB threshold	-300 mV (288 to 312 mV)
Dropout	214 mV sim / 330 mV max at 200 mA
Current limit	126.6 mA programmed; internal backup 220 mA min
GND pin current	2.35 mA quiescent
PSRR at 1.2 MHz	-77.9 dB (simulated)
Firmware delay	Wait $\geq$ 170 ms after PG before enabling the OPA1622
PG at cold start	Reads high until IN passes $\approx$ -1.8 V; ignore PG until +3V3 is up
PG in shutdown	Pulled low: reads not-good (opposite of the LT3042); 33 μ A through R_PU
C_OUT requirement	$\geq$ 4.7 μ F, ESR < 30 m Ω , ESL < 1.5 nH
θ_{JA} (DFN)	34 $^{\circ}$ C/W

Source: [1]

7. Open Problems

#	Problem
1	Exposed pad is IN (-3V9): dedicated pour and via field; the footprint must not tie it to GND.
2	PG brownout warning is 2.2 ms at the simulated ramp rate; a real brownout is faster. Consider a hardware mute from PG.
3	ILIM scale-factor tolerance: confirm the minimum limit stays above 98 mA.
4	Remaining simulations: current limit, PG in shutdown, full chain with the LT3462A.

8. Revision Log

Rev	Date	Change
0 to 4	2026-09-22	Initial issue; PG divider 135 k Ω / 15 k Ω ; simulation results.
5, 6	2026-09-22	Maximum output 0.6 V RMS; peak load 98 mA; current limit 125 mA (R_ILIM 15.6 k Ω).
7, 8	2026-09-23	Input from the LT3462A inverter; C_SET 2.2 μ F tantalum, cathode to SET.
9	2026-09-27	Reformatted.
10	2026-10-03	Reviewed against the schematic. R_ILIM 15.4 k Ω (#1-088): 126.6 mA. R_PG1 0603 (#1-086), R_EN2 1 % (#1-087). R_PU 100 k Ω (#1-131): 33 μ A, within the PG specification. Designators added. Closed: DAC supply allocation (dedicated LT3042), package, R_PU, part numbers.

9. Sources

1. LT3093 Datasheet (Analog Devices)
2. DAP schematic sheet DAP-Power-2, -3.3 V LDO (author, 2026-10-03)
3. Alex-Loker-Altium-Database.xlsx, DAP rows (author)